EXP NO: 07

SLIDING WINDOW PROTOCOL

Aim:

Write a program to implement flow chart of data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another

CODE:

Sender:

```
import time
import os
window_size = int(input("Enter window size: "))
message = input("Enter the message: ")
frames = []
for i, ch in enumerate(message, start=1):
    frame.append((i, ch))
sender_buffer = "sender_buffer.txt"
receiver_buffer = "Receiver_buffer.txt"
start=0
while start < len(frames):
    window = frames[start:start+window_size]
    print("\nSender")
    with open(sender_buffer, "w") as file:
        for frame_no, data in window:
            file.write(f"{frame_no},{data}\n")
            print("Frame", frame_no, ":", data)
print("Frames written to sender_buffer.txt")
input("\nPress enter to continue.")
if not os.path.exists(receiver_buffer):
    print("Receiver_buffer.txt not found")
    continue
```

```
        with open(receiver_buffer,"r") as file:
            response = file.read().strip()
        print("Received:", response)
        if response.starts with("Ack:"):
            ack_no = int(response.split(":")[1])
            expected_ack = window[-1][0]
            if ack_no == expected_ack:
                print("Ack", ack_no, "received")
                print("Window moved")
                start = start + window_size
            else:
                print("Wrong Ack")
                print("Expected:", expected_ack)
                print("received", ack_no)
                print("Resending frames")
        elif response.starts_with("NACK:"):
            nack_no = int(response.split(":")[1])
            print("NACK", nack_no, "received")
            print("Resending from frames", nack_no)
            start = nack_no - 1
        else:
            print("Invalid response")
        time.sleep(1)
    print("\n All frames are sent")
```

Receiver:

```
import os
sender_buffer = "Sender_Buffer.txt"
received_buffer = "Received-Buffer.txt"
```

```
expected_frame = int(input("Enter expected frame no:"))
print("In Receiver")
if not os.path.exists(sender_buffer):
    print("sender_Buffer.txt not found")
else:
    with open(sender_buffer, "r") as file:
        line = file.readlines()
    error = False
    last_frame = expected_frame_1
    print("Frames. received")
    for line in lines:
        line = line.strip()
        if line == "":
            continue

    parts = line.split(",")
    frame_no = int(parts[0])
    data = parts[1]
    print("Frame" + frame_no, ":", data)
    If frame_no == expected_frame:
        print("Frame", frame_no, "received correctly")
        last_frame = frame_no
        expected_frame += 1

    else:
        print("Error in frames", frame_no)
        print("Expected frames:", expected_frame)
        error = True
        with open(receiver_buffer, "w") as file:
            file.write("NACK: {expected_frame}")
        print("NACK", expected_frame, "sent")
        break.
```

```
if not error:
    with open(receiver_buffer, "w") as file:
        file.write(f"ACK:{last_frame}")
    print("ACK", last_frame, "sent")
```

Output:

Sender (window 1):

Enter window size: 3
Enter the message: HELLO
Sender
Frame 1: H
Frame 2: E
Frame 3: L
Frames written to sender_Buffer.txt
Received: ACK:3
Ack 3 received
window moved.
Sender
Frame 4: L
Frame 5: O
Frame written to sender_Buffer.txt
Received: ACK:5
ACK 5 received
Window moved.
All frames sent successfully

Receiver (Window 2):

Expected frame no: 1
Receiver
Frame 1: H
received correctly
Frame 2: E

received correctly
Frame 3: 4
received correctly
ACK 3 sent
Expected frame no: 4
Receiver:
Frame 4:
received correctly
Frame 5:
received correctly

Result:

Thus, the frames are sent and received using the sliding window protocol.